

Protocol for
Falls: a marker of preclinical Alzheimer's disease

Stark, Susan*, Ances, Beau M., Xiong, Chengjie, Balota, David, Bollinger, Rebecca, Somerville, Emily, Thompson, Regina, Grant, Elizabeth, Morris, John

*To whom correspondence should be addressed

Washington University School of Medicine
5232 Oakland Ave
St. Louis, MO 63110

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Protocol modification log

Date	Version	Description
12.26.18	1.1	Modified personnel and updated assessments
01.16.19	1.2	Updated personnel and assessments
02.15.19	1.3	Modified exclusion criteria; added phone option for some assessments
03.05.19	1.4	Addition of sensation assessment
05.08.19	1.5	Addition of optional stool sample collection
07.30.19	1.6	Addition of UPSIT test and re-consent item
11.22.19	1.7	Addition of dietary questionnaire and GI questionnaire for stool sample assessment; updated personnel
02.28.20	1.8	Addition of hearing screen, questionnaires, and updated personnel
03.24.20	1.9	Addition of coronavirus (COVID-19) changes
04.17.20	1.10	Added telehealth and coronavirus (COVID-19) assessments
05.20.20	1.11	Added procedures to re-consent and mail stool kits during coronavirus pandemic
07.28.20	1.12	Added consent via REDCap and edited inclusion criteria
10.27.20	1.13	Updated formatting, personnel, and references
01.26.21	1.14	Updated stool collection portion only
03.17.21	1.15	Updated remote/telehealth procedures, COVID-19 diagnoses and symptoms
06.01.21	1.16	Updated stool collection portion for olfaction
10.08.21	1.17	Updated personnel
01.25.22	1.18	Addition of MoCA & AD8
02.08.22	1.19	Updated stool procedures
02.16.22	1.20	Updated staff roles and responsibilities
06.09.22	1.21	Addition of DS supplement
07.15.22	1.22	Addition of social media as a recruitment source for DS supplement
07.27.22	1.23	Addition of EPIC as recruitment source for DS supplement
08.03.22	1.24	Addition of Volunteer for Health as recruitment source for DS supplement
10.31.22	1.25	Modified number of monthly fall calls for DS supplement
12.08.22	1.26	Modified length of time for monthly fall calls for DS supplement
05.12.23	1.27	Updated MoCA Information
06.02.23	1.28	Updated AD8 and MoCA Information
08.04.23	1.29	Modified DS supplement recruitment sources
10.20.23	1.30	Added information about sharing de-identified data for DS supplement
10.31.23	1.31	Added additional information about sharing de-identified data for DS supplement
12.09.24	1.32	Addition of Opal wearable sensors for select in-home assessments

Co-Authors:**Susan Stark, PhD, OTR/L**

Program in Occupational Therapy
Washington University School of Medicine
St. Louis, Missouri

Beau M. Ances, MD, PhD, MSc

Department of Neurology
Washington University School of Medicine
St. Louis, Missouri

Chengjie Xiong

Division of Biostatistics
Washington University School of Medicine
St. Louis, Missouri

David Balota, PhD

Department of Psychological & Brain Sciences and Neurology
Washington University School of Medicine
St. Louis, Missouri

John Morris, MD

Department of Neurology
Washington University School of Medicine
St. Louis, Missouri

Rebecca Bollinger, OTD, OTR/L

Program in Occupational Therapy
Washington University School of Medicine
St. Louis, Missouri

Regina Thompson

Department of Neurology
Washington University School of Medicine
St. Louis, Missouri

Elizabeth Grant, PhD

Division of Biostatistics
Washington University School of Medicine
St. Louis, Missouri

Selena E. Washington, PhD, MSPH, OTR/L (DS supplement)

Department of Occupational Science and Occupational Therapy
Saint Louis University
St. Louis, Missouri

WORLD HEALTH ORGANIZATION DATA SET

Primary Registry and Trial Identifying Number	N/A
Date of Registration	N/A
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Source(s) of Monetary Support	National Institute on Aging Paula and Rodger Riney Fund (stool portion only)
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Contact for Public Queries	Susan Stark, PhD, OTR/L ssstark@wustl.edu 314-273-4114 Department of Occupational Therapy Washington University School of Medicine 5232 Oakland Ave St. Louis, MO 63110 United States
Contact for Scientific Queries	Susan Stark, PhD, OTR/L ssstark@wustl.edu 314-273-4114 Department of Occupational Therapy Washington University School of Medicine 5232 Oakland Ave St. Louis, MO 63110 United States
Public Title	Falls: a marker of preclinical Alzheimer's disease
Scientific Title	Falls: a marker of preclinical Alzheimer's disease
Countries of Recruitment	United States
Health Condition(s) or Problem(s) Studied	Falls and preclinical AD
Intervention(s)	N/A
Key Inclusion and Exclusion Criteria	Participants will include cognitively normal (CN) older adults currently enrolled in longitudinal studies of memory and aging at the Knight ADRC. Participants are community-dwelling individuals recruited from the greater St. Louis, MO., metropolitan area. Individuals with a medical or psychiatric illness that could interfere with longitudinal follow-up, prevent participation in neuroimaging, or adversely affect cognition are excluded.

Study Type	Longitudinal cohort study Assignment: N/A Primary purpose: To examine the relationship between falls and functional mobility in preclinical stages of Alzheimer disease (AD)
Date of First Enrollment	March 2019
Target Sample Size	350
Recruitment Status	Enrolling
Primary Outcome(s)	Aim 1: To examine the relationship between falls and functional mobility in preclinical stages of AD. a) We hypothesize that CN individuals in preclinical Stages 2 and 3 will have more falls (number and severity) and more impaired functional mobility compared to CN individuals in Stages 0 and 1. We propose that assessment of falls and functional mobility will provide a “real-life” evaluation of preclinical AD. b) We hypothesize that the rate of change in falls and impaired functional mobility will be greater in CN older adults who subsequently develop symptomatic AD compared to CN individuals who do not develop AD. Aim 2: To examine a hypothesized model of central and peripheral mechanism(s) underlying falls and functional mobility in preclinical stages of AD. a) We hypothesize that a loss of connections within and between brain networks underlies declines in the central (neuropsychological performance) and peripheral (balance, strength, vision, and sensation) nervous system changes that occur in preclinical AD. b) We hypothesize that an interaction of both central and peripheral mechanisms will underlie a loss of falls and functional mobility.

ORGANIZATIONAL STRUCTURE AND RESPONSIBILITIES

Susan Stark, PhD, MSOT, OTR/L, Principal Investigator

Dr. Stark will jointly oversee the activities of this project with Dr. Ances. She will explore the transition of the role of falls in asymptomatic, preclinical stages of Alzheimer disease (AD; Aim 1). Dr. Stark will manage all recruitment and retention, all personnel in the field, and the prospective collection of fall data. She will conduct staff training, weekly staff meetings, and quarterly investigator meetings. Dr. Stark will draft manuscripts and present data at national and international conferences.

Dr. Stark will serve as contact PI and will assume fiscal and administrative management including maintaining communication with other team members through monthly meetings. She will directly supervise Dr. Bollinger and a post-doctoral associate to be named. She will work with Dr. Ances to coordinate all aspects of the study including analysis of data and generation of manuscripts.

Beau M. Ances, MD, PhD, MSc, Principal Investigator

Dr. Ances will serve as the chair of the steering committee. He will directly supervise research coordinators and will work with post-doctoral fellows in the analysis of neuroimaging data.

Chengie Xiong, PhD, Co-Investigator

Dr. Xiong oversees all of the data from the longitudinal studies that we will access for the analyses we have planned for this study. Dr. Xiong will supervise all statistical analyses for this study. He will review and supervise new data-entry screen development for this study. He will serve on the steering committee and will contribute to manuscripts, presentations, and preparation of a subsequent application. Dr. Xiong will also supervise the work of data manager Dr. Grant.

David Balota, PhD, Co-Investigator

Dr. Balota will supervise the collection of all neuropsychological data by Research Coordinator Ms. Thompson. He will interpret cross-sectional and longitudinal neuropsychological evaluations of all of the participants who will be recruited for this study. Dr. Balota will contribute to manuscripts, presentations, and preparation of a subsequent application.

John Morris, MD, Co-Investigator

As the leader of the Knight ADRC, he will facilitate access to study participants and integrate the study into ongoing workflow and procedures to ensure seamless integration of the proposed study into ongoing studies. He will contribute to manuscripts, presentations, and preparation of a subsequent application. He will serve on the steering committee and will act as arbitrator as needed.

Rebecca Bollinger, OTD, OTR/L, Study Coordinator

Dr. Bollinger will coordinate all aspects of the research project, including scheduling, managing visits and mailings, reviewing data files for completeness, and maintaining consent forms and institutional review board approval. She will process all participant payments. She will work closely with both PIs and Knight ADRC staff to identify potential participants and will coordinate a consent visit in conjunction with the potential participant's annual assessment. She will serve as point of contact for participants monthly (for reporting falls) and will employ strategies to improve retention (e.g., mailing birthday cards). She will attend quarterly meetings at the Knight ADRC (required for all studies accessing Knight ADRC participants). She will schedule and coordinate all in-home visits. She will complete home visits. She will create, schedule, and coordinate the distribution of fall calendars. She will schedule and conduct all follow-up interviews. Dr. Bollinger will work closely with Regina Thompson and the Knight ADRC staff to ensure that the study is integrated and seamless for participants. Under the guidance of Dr. Grant, Dr. Bollinger will be responsible for entering the in-home assessment results into a REDCap database and for recording the visit in the Knight ADRC REDCap database. She will meet weekly with Drs. Ances, Stark, Balota, and Ms. Thompson to ensure that project deadlines are met.

Regina Thompson, Clinical Research Coordinator II

She will assist Dr. Bollinger in identifying and recruiting potential participants seen at the Knight ADRC. Ms. Thompson will work in tandem with Dr. Bollinger to obtain written informed consent from participants. Along with Dr. Bollinger and under the guidance of Dr. Grant, Ms. Thompson will enter neuropsychological data into the database. She will maintain regular communication with Dr. Ances to ensure the proper and timely collection, upload, and quality control of all neuroimaging data. She will serve as a key liaison among the neuroimaging, occupational therapy, and biostatistics teams that will contribute to this project. She will meet weekly with Drs. Ances, Stark, and Balota as well as Drs. Bollinger to ensure that project deadlines are met, and she will serve as a liaison among all study staff and the Steering Committee.

Occupational Therapists/Occupational Therapy Assistants:

Debbie Lewis, MOT, OTR/L

Audrey Keleman, OTR/L

Meghan Haxton, COTA/L

Emily Somerville, OTD, OTR/L

They will conduct home visits as needed under the direction of Dr. Bollinger.

Graduate Assistants:

They will assist with stool sample collection and data analysis under the direction of Ms. Thompson and Dr. Bollinger.

Data Management:

Elizabeth Grant, PhD, Research Statistician

Responsibilities include: Statistical design of study; data verification; developing and implementing data management procedures including the data flow and procedures for data entry, error identification, and correction; and preparing quarterly reports on enrollment, participant status (e.g., withdrawals), adverse events, and independent safety monitoring body reports.

Stool Analysis:

Ibrahima Ndao, MD, Clinical Research Specialist

Carla Hall-Moore, BS, Research Lab Manager

Kimberley Sukhum, PhD, Postdoctoral Research Associate

Aura Ferreiro, BS, Graduate Research Assistant

Responsibilities include: Collection, processing, and analysis of optional annual stool samples.

DS supplement:

Selena Washington, MPH, PhD, OTR/L, Trainee

For the DS supplement, Dr. Washington will manage all recruitment and retention and the prospective collection of fall data. She will schedule and complete all home visits, distribute calendars, and serve as point of contact for participants to report falls monthly. Dr. Washington will continue to work closely with the parent study staff (Stark, Ances, and Bollinger) and the ABC-DS study/Ances clinical staff (Ances, Westerhaus, and Nelson). She will draft manuscripts and present outcomes at local, national, and international conferences.

Abstract

Conversion to symptomatic Alzheimer disease (AD) is not an immediate process; rather, it develops slowly over a series of preclinical stages. Although symptomatic AD is characterized by progressive cognitive problems, growing evidence suggests that functional mobility abnormalities and an increase in falls may precede cognitive impairment. Falls are a leading cause of morbidity and mortality in the older adult population. Older AD patients have more than twice the risk of falls compared to the general population. Measures of everyday function are currently not included in the evaluation of cognitively normal individuals with preclinical AD. Within a well-characterized cohort of older, cognitively normal individuals (≥ 65 years old) who are followed by the Knight Alzheimer's Disease Research Center, we will perform longitudinal in-home assessments to quantitatively evaluate falls and functional mobility. We hypothesize that brain neuropathology (e.g., the presence of amyloid and/or tau proteins, evidence of neurodegeneration) lead to changes in functional connectivity of brain networks. These brain network changes affect both central nervous system markers (e.g., cognitive performance) and peripheral nervous system markers (e.g., balance, strength, vision, and sensation). We suggest that a synergistic interaction between central and peripheral changes leads to altered functional mobility, resulting in an increase in falls. Our findings will advance the field of AD by identifying novel "real-world" biomarkers in preclinical stages of AD. Assessment of falls and gait instability in the home setting could potentially serve as an inexpensive and non-invasive screening tool to identify individuals at greatest risk for conversion to symptomatic AD. This may have important implications for the timing of interventions in secondary prevention trials in AD.

DS supplement:

Adults aging with Down syndrome (DS) are at high risk for the development of Alzheimer's disease (AD).¹ Nearly all adults with DS develop the neuropathological features of preclinical AD by the age of 40 years.¹ Similar to the cognitively normal adult population, there is also an urgent need to understand the preclinical and early phases of AD progression within the adult population aging with DS. It is hypothesized that brain neuropathology (e.g., the presence of amyloid, tau proteins, and neurodegeneration) leads to changes in functional connectivity of brain networks and that a synergistic interaction between central and peripheral changes leads to altered functional mobility, resulting in an increase in falls.² While individuals with DS have a higher rate of falls, it remains unknown when falls occur in the progression to AD within adults with DS.^{3,4} There is a need to expand the research to include functional measures such as falls, fall risk, and functional mobility.⁵ Falls and functional mobility are associated with clinical measures of cognition and may be a non-cognitive behavioral biomarker of AD.^{3,6,7} A better understanding of function could inform this biomarker profile through a direct and less invasive method of assessment, therefore reducing burden for both the participant and caregivers.^{1,3,8}

Currently, the Alzheimer's Biomarkers Consortium — Down Syndrome (ABC-DS), of which WUSM is a site, is contributing to the largest U.S.-based, multisite, biomarker initiative to target preclinical AD in the DS population. The measures collected by the consortium include clinical, cognitive, and AD measures.³ This study will follow a cohort of community-dwelling adults with DS currently participating in the ABC-DS study at WUSM, and aging adults with DS within the greater St. Louis metropolitan area. It will serve as the necessary groundwork for a larger study related to falls and preclinical AD. This study will quantitatively examine falls and functional mobility among adults aging with DS and their relationship with preclinical AD; these measures could potentially serve as a method to screen people aging with DS for AD pathology and lead to a better understanding of functional and cognitive changes that occur within this population.

Background

This proposal will systematically characterize cerebral network dysfunction, functional mobility, and falls in a cohort of cognitively normal (CN) older adults (≥ 65 years old) with and without preclinical Alzheimer disease (AD). In doing so, we explore the novel hypothesis that falls are an indicator of preclinical AD. Falls are the leading cause of injury, long-term disability, premature institutionalization, and injury related-death in older individuals.⁹⁻¹² AD patients have a 60%-80% increased risk of falls.¹³ This risk is more than twice that of the general population.¹⁴ Functional mobility, or the ability to navigate safely in the real world, is a complex task that requires an interplay amongst attention, executive, motor, and sensory processing. Declining functional mobility (gait and balance) and consequential falls¹⁵ may be a bellwether of loss of integration in the central and peripheral nervous systems.^{16,17}

Conversion from preclinical to symptomatic AD occurs over decades.¹⁸ Changes in molecular biomarkers can now be identified as an individual transitions from CN to preclinical AD to symptomatic AD.¹⁹ Currently, criteria for preclinical AD stages classify CN individuals by neuroimaging and/or cerebral spinal fluid (CSF) measures. CN Stage 0 individuals have no biomarker abnormalities. CN Stage 1 individuals have amyloidosis. CN Stage 2 individuals have both amyloidosis and neurodegeneration. CN Stage 3 individuals have evidence of subclinical cognitive changes.²⁰⁻²³ Measures of everyday function are currently not included in the evaluation of CN individuals with preclinical AD. Growing evidence suggests functional mobility changes occur in symptomatic AD, but gaps exist as to whether functional mobility and falls are a preclinical marker of AD.²⁴ Our preliminary data suggest that falls occur at higher rates in the preclinical phase of the disease and that the mechanisms that underlie deteriorating cognitive function are associated with declines in gait and balance. In this proposal, we will identify the preclinical stage when falls begin to occur.

Novel, non-invasive neuroimaging methods can assess functional connections within the brain as an individual quietly rests in a scanner.²⁵ Resting state functional connectivity (rs-fc) decreases in symptomatic AD, and our preliminary data demonstrate that it is also decreased in CN individuals with preclinical AD. Specifically, we have identified that a reduction in rs-fc occurs in the dorsal attention network (DAN), a set of regions involved in attentional control.²⁶ Connections within the DAN and between other brain networks may lead to an inability to process the complex demands to navigate an environment. The relationship between rs-fc and functional mobility and falls, especially in CN individuals with preclinical AD, is unknown.

The team will study an existing cohort of CN older adults that has been well characterized using multiple modalities (including neuropsychological performance testing, clinical evaluation, CSF, and neuroimaging) in the Knight Alzheimer's Disease Research Center (ADRC). Our team will generate a unique set of functional and fall-related variables in participants 65 years or older with and without preclinical markers of AD. We have scientifically justified our study based on the following preliminary data.

CN older adults with preclinical AD have a higher risk of falls compared to CN older adults without preclinical AD.

To examine the relationship between preclinical AD and falls, 125 CN older adults from the Knight ADRC prospectively reported falls monthly using a calendar journal to evaluate the occurrence of falls in the home and community.²⁷ Falls were confirmed by a standardized interview. Participants were divided into those with the most extreme values of amyloidosis using positron emission tomography (PET) imaging (radiotracer [¹¹C]–Pittsburgh Compound B [PiB]). Overall, the sample was predominately female (74.4%) and white (78.8%), with a mean age of 74.5 years old. The 125 participants reported 154 falls over the 365-day monitoring period; 75 individuals (66%) had at least one fall over the 1-year interval, with the total number of falls for a participant ranging from 0–12. The cumulative incidence of first fall was greater in the preclinical AD group after controlling for activities of daily living (ADLs; hazard ratio [HR], 2.95; 95% confidence interval [CI], 1.01–6.45; $p = 0.05$; Figure 1). This promising preliminary data did not control for known fall risk factors and

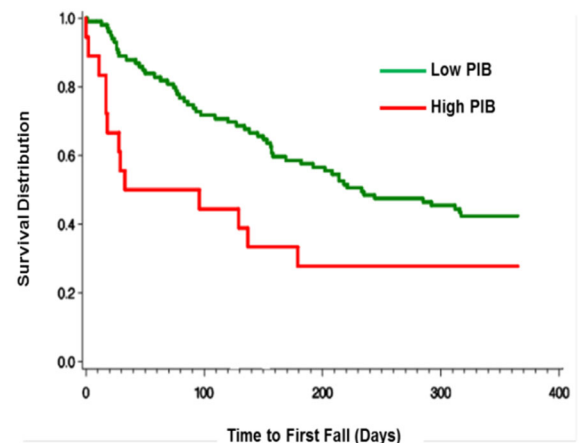


Figure 1. Kaplan-Meier curves for time to first fall for CN with preclinical AD and CN without preclinical AD participants. PIB, Pittsburgh Compound B.

was not sufficiently powered to examine the rate of falls by preclinical stage. The proposed study will extend these findings by increasing the sample size, controlling for fall covariates, and using molecular markers to stage preclinical AD.

Table 1. Multiple linear regression of CSF biomarkers predicting falls and severity of falls at 3 years

n = 34	Variable	Standard Coefficient	t	CI
No. of Falls	(Constant)		1.70	-0.45~4.87
	Age	-0.28	2.24	0.03~0.63
	Gender	0.34	-1.69	-0.06~0.01
	A β_{42}	-0.40	-2.47	-1.43E-3~-1.40E-4
Severity of Falls	(Constant)		1.87	-0.38 ~ 8.72
	Age	-0.31	-1.90	-0.11 ~ 0.004
	Gender	0.32	2.06	0.01 ~ 1.03
	A β_{42}	-0.37	-2.25	-0.002 ~ -1.1E-4

Longitudinal changes in preclinical AD biomarkers are related to the number and severity of falls among CN older adults. We performed a 3-year retrospective analysis of the number and severity of falls from a cohort of 97 CN older adults followed at the Knight ADRC (mean age, 74.2 $\pm$ 6.5 years old; 66% female). Falls were reported at annual interviews. A multiple linear regression analysis explored the rate and severity of falls as a function of CSF molecular biomarkers (A β_{42} , tau, and p-tau) while controlling for age and gender. Table 1 shows that A β_{42} was a significant predictor of both frequency of falls ($F = 4.58$, $p = .01$, $R = .57$, $R^2 = .33$) and fall severity ($F = 4.15$, $p = .02$, $R = .55$, $R^2 = .31$) at year 3 ($n = 34$). These results suggest that longitudinal changes in falls are sensitive in detecting preclinical AD changes. The current proposal will extend these preliminary findings by increasing

the sample size, controlling for important fall covariates, and using molecular markers to longitudinally assess as CN participants progress to symptomatic AD.

Molecular biomarkers can stage preclinical AD.²² We examined data from CN participants enrolled in

longitudinal aging studies at the Knight ADRC. Participants were classified by CSF A β_{1-42} and tau concentrations and a memory composite score as either normal, preclinical AD Stage 1–3, or suspected non-AD pathophysiology (SNAP; abnormal injury marker without abnormal amyloid marker). Of the 311 participants, 129 (41%) were classified as normal, 47 (15%) as Stage 1, 36 (12%) as Stage 2, 13 (4%) as Stage 3, 72 (23%) as SNAP, and 14 (5%) remained unclassified. The 5-year progression rate to a Clinical Dementia Rating (CDR) of 0.5 or more (symptomatic AD) was 2% for participants classified as normal, 11% for Stage 1, 26% for Stage 2, and 56% for Stage 3 (Figure 2.) There is a gap in understanding where changes in functional mobility and falls lie within preclinical AD stages and the progression rate for CN individuals to symptomatic AD. Within a well-characterized, longitudinally assessed cohort, we will extend these findings by including fall risk within preclinical stages of AD.

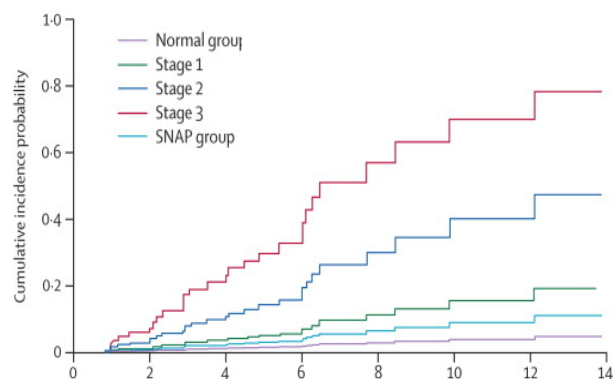
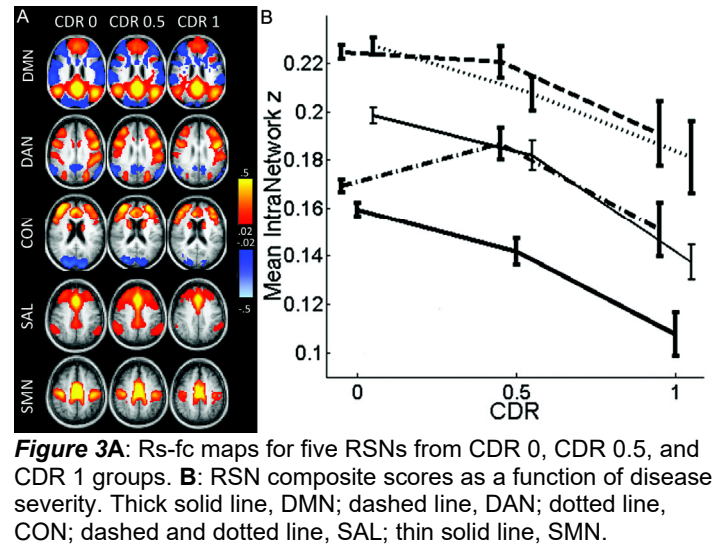


Figure 2. Progression to symptomatic AD (CDR > 0) from preclinical AD stage over 14-year follow-up.

Mild changes in functional mobility, as captured by performance-based assessments, predate conversion to symptomatic AD. A cohort of 435 CN adults 60 years or older underwent annual assessments of physical and cognitive function at the Knight ADRC over a 5-year interval. A subset of these participants ($n = 81$) subsequently developed symptomatic AD over this period. Those who developed AD were older (81.0 vs. 74.2 years old, $p = 0.001$) and had worse performance on the Physical Performance Test (PPT; 25.5 vs. 28.1; $p = .009$), which is a clinical performance-based measure of strength, mobility, dexterity, and stamina. In an adjusted analysis (age, sex, education, and APOE $\epsilon 4$ allele), the nine-item PPT scores significantly predicted time to diagnosis (HR, 0.94; 95% CI, 0.89-0.99; $p = .02$). These results suggest that mild functional mobility impairments are associated with subsequent development of symptomatic AD. Within a well-characterized, longitudinally assessed cohort, we will examine fall risk within each of these preclinical stages of AD.

Changes in rs-fc occur with symptomatic AD, mild cognitive impairment, and mild AD. Multiple resting state networks (RSNs) were assessed among CN (CDR 0, $n = 386$), MCI (CDR 0.5, $n = 91$), and mild AD

(CDR 1, $n = 33$) including the DMN, DAN, control (CON), SAL, and sensorimotor (SMN; Figure 3A).²⁸ We used 36 seed regions. With increasing disease progression, both intra- and inter-network functional correlations were disrupted. By symptomatic AD (CDR 1), the integrity of all intra-network RSNs was significantly diminished compared to CDR 0. Intra-network connections within the DMN, DAN, and CON were diminished for MCI (CDR 0.5) compared to CN (CDR 0) participants. Intra-network functional correlations within the SAL were increased for MCI (CDR 0.5) compared to CN (CDR 0) individuals (Figure 3B). Similar results were also seen for inter-network relationships, with a loss in connections between networks with increasing degree of cognitive impairment. This proposal will study the effects of network dysfunction on downstream measures of functional mobility and falls.



Rs-fc is decreased for CN older adults with falls compared to CN older adults without falls. Rs-fc was performed on a sample of 24 CN older adults with falls (mean age, 75 years old) and 28 CN older adults without falls (mean age, 71 years old). Two hundred ninety-eight regions, representing 13 RSNs, were derived by maximizing the topographic concordance between results obtained by seed-based correlation mapping. To assess the changes in all seeds for all RSNs, matrices were computed (Figure 4). Intra-network correlations are displayed in blocks along the diagonal; cross-network correlations appear in off-diagonal blocks. The CN older adults with falls (middle) and no falls (right) matrices show positive correlations within each RSN and negative correlations between networks. The difference matrix (Falls–No Falls) was also computed (left). CN older adults with falls had significant decreases in intra-network rs-fc within the DAN, DMN, and visual (VIS) networks. Inter-network functional connections between the DAN and DMN were diminished in CN fallers compared to CN non-fallers. This proposal will study whether changes within and between RSNs lead to falls and gait instability prior to subjective memory complaints.

Error rates for the Stroop task are a marker for conversion to symptomatic AD. We completed a retrospective analysis of a sample of 47 CN older adults to explore the utility of the computerized Stroop task, which measures attention and

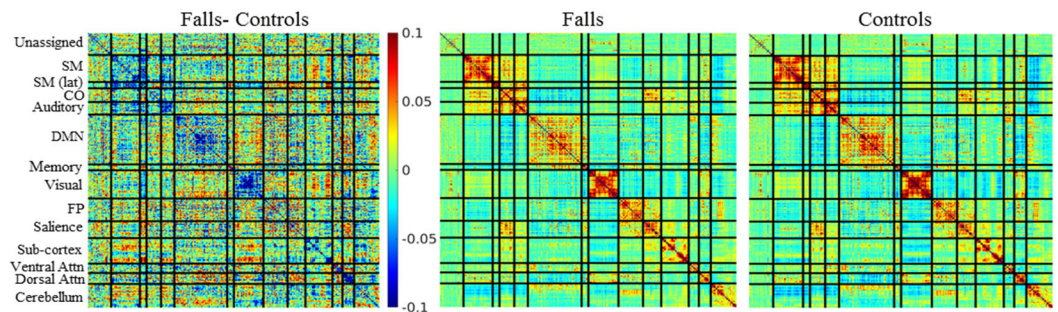


Figure 4. ROI pair correlation matrices for falls–no falls (left), falls (middle), no falls (right) CN individuals

processing speed, to discriminate those who convert to AD from those who do not. Errors on color incongruent trials and changes in the tail of the reaction time distribution were the strongest predictors of conversion to symptomatic AD. Differences were not observed for standard episodic memory measures, logical memory, or associated recall. We developed an attentional composite (including error rates from the Stroop, Simon, and Attention Switching tasks) to examine the relation between attentional control and biomarkers in a sample of 238 CN individuals. This attentional composite was sensitive to differences in CSF A β_{42} at baseline and longitudinally. Preclinical individuals with high CSF tau exhibited a steeper decline in the attentional control composite than individuals low in CSF tau, suggesting sensitivity between our cognitive measures and preclinical biomarkers.²⁹

RSN composite scores are associated with Stroop performance. We have assessed the relationship between reaction time distributional analysis, error rates from the Stroop task, and rs-fc in four distinct RSNs: the DMN, SAL, DAN, and SMN, within a sample of 237 CN older adults. A larger Stroop effect in the error rate in the incongruent condition was associated with reduced rs-fc in both the SAL and DMN. The slow tail of the reaction time distribution for the Stroop task was also related to reduced rs-fc within the SAL. These results suggest that a link exists between RSN composite scores and specific attentional performance measures.³⁰ This proposal will examine the relationship between attentional/executive mechanisms and their underlying RSNs to the occurrence of falls.

DS supplement:

Nearly all adults with DS develop the neuropathological features of preclinical AD by the age of 40 years.³ Similar to the cognitively normal (CN) adult population, there is also an urgent need to understand the preclinical and early phases of AD progression within the adult population aging with DS. While individuals with DS have a higher rate of falls, it remains unknown when falls occur in the progression to AD within adults with DS.^{6,7} Currently, the Alzheimer's Biomarkers Consortium—Down Syndrome (ABC-DS), of which WUSM is a site, is contributing to the largest U.S.-based, multisite biomarker initiative to target preclinical AD in the DS population. The measures collected by the consortium include clinical, cognitive, and AD measures.³ There is a need to expand this research to include functional measures such as falls, fall risk, and functional mobility.³ Falls and functional mobility are associated with clinical measures of cognition and may be a non-cognitive behavioral biomarker of AD.^{8,31,32} A better understanding of function could inform this biomarker profile through a direct and less invasive method of assessment, therefore reducing burden for both the participant and caregivers.^{3,33,34}

Dr. Washington will be guided by mentors Drs. Stark and Ances, who are in collaboration regarding a similar area in late-onset Alzheimer disease (LOAD). Dr. Ances is the site PI for the ABC-DS. Drs. Stark and Ances' expertise in aging will support Dr. Washington's understanding of key pathophysiological changes seen in neurodegenerative diseases. Dr. Washington will follow a cohort of community-dwelling adults with DS currently participating in the ABC-DS study at WUSM. This study will examine falls and functional mobility among adults aging with DS and their relationship with preclinical AD. It will serve as the necessary groundwork for a larger study related to falls and preclinical AD. This is the first study, to our knowledge, to quantitatively assess patterns of falls and functional mobility impairments that occur within a community-dwelling adult population aging with DS. The central hypotheses are: (1) similar to older adult populations, falls and impaired functional mobility in aging adults with DS may represent a sign of preclinical AD, and (2) implementation of evidence-based assessments and measures will expand our knowledge of falls among adults with DS and their relationship with preclinical AD. These measures could potentially serve as a method to screen people aging with DS for AD pathology and will lead to a better understanding of functional and cognitive changes that occur in this population.^{2,35,36}

Study Design

All study aims will involve cross-sectional and longitudinal data from a cohort of CN community-dwelling older adults (Figure 5). We will leverage the Knight ADRC infrastructure to recruit and enroll a sample of 350 older adults (≥65 years old) and follow them longitudinally for 4 years. We will benefit from the existing Knight ADRC infrastructure, including a well-characterized cohort of CN older adults with longitudinal clinical and neuropsychological assessment and biomarker studies of biofluids (blood, CSF) and neuroimaging (amyloid PET, structural and functional MRI; green boxes in Figure 5). All Knight ADRC enrollees agree in principle to

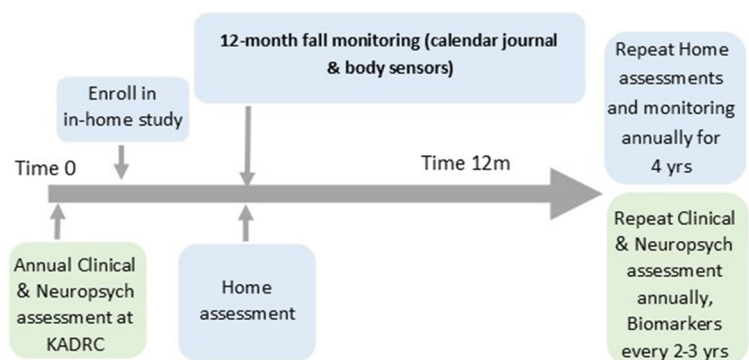


Figure 5. Research Design overview. Data that will be collected annually include in-home assessments and prospective fall reporting using calendars and functional mobility measurements (blue boxes). Participants will be followed for 4 years. Green boxes are measures of interest that are currently collected by the Knight ADRC (no cost).

participate in all biomarker procedures. We will collect additional data (blue boxes in Figure 5) including non-cognitive behavioral characteristics and environmental factors. By leveraging the vast Knight ADRC resources, we can accomplish more than would be possible from a single funding source. As noted in the preliminary data section, there is high enthusiasm by participants to have home monitoring, and few participants have been lost to follow-up. In the interest of space, new manipulations are described in detail in this application, but standardized procedures that are well established at the Knight ADRC are referenced and briefly described. Throughout the application, we have attempted to use robust and unbiased strategies to assess falls and functional mobility during the progression to symptomatic AD.

DS supplement:

We will leverage the ABC-DS infrastructure to recruit and enroll a sample of 30 aging adults (≥ 25 years old) with Down syndrome and follow them over a course of 3-9 months. We will utilize the existing ABC-DS and Knight ADRC infrastructure during the implementation of this study to expand our research efforts toward recruitment and enhance the scientific rigor of this study.

Study Participants

Participants will include CN older adults currently enrolled in longitudinal studies of memory and aging at the Knight ADRC^{37,38}. Inclusion criteria: (1) 65 years or older and (2) CDR³⁹ 0 at their most recent clinical assessment, indicating cognitive normality, or if a CDR isn't available, AD8 ≤ 3 and MoCA score of ≥ 23 .

Exclusion criteria: History of Parkinson's disease. The Unified Parkinson's disease Rating Scale (UPDRS) will be used to assess the course or progression of Parkinson's disease.⁴⁰

In order to acquire a more robust sample and specifically characterize differences in immune dysfunction effects in the stool or intestinal mucosal microbiome/microbiota/metagenome in symptomatic vs. non-symptomatic individuals, and olfaction, participants only able to complete the stool collection portion of the study may be CDR ≤ 2 and ≥ 50 years of age.

DS supplement:

Participants will include aging adults with Down syndrome enrolled in the ABC-DS study at the WUSM; and adults with DS recruited from the greater St. Louis, MO area. Inclusion criteria: community dwelling adults with DS (full trisomy, partial trisomy, or mosaic Down syndrome), age ≥ 25 years, cognitively stable or mild cognitive impairment as indicated by a score of < 20 on the Dementia Screening Questionnaire for Individuals with Intellectual Disabilities (DSQIID), and willingness to participate in study procedures and assessments; Exclusion criteria: a condition(s) that would interfere with cognitive assessments including: unstable medical or a psychiatric condition, severe motor or sensory impairment, and/or IQ less than 30-according to the KBIT or from other medical/clinical documentation to verify an IQ of 30.

Recruitment

At the time of their annual clinical assessment, we will invite individuals to participate in this study if they meet the inclusion criteria. Based on the Knight ADRC historic rate of conversion to symptomatic AD, over the next 4 years, we conservatively estimate that 28.6% of the sample will progress from CN (CDR 0) to MCI or AD (CDR > 0). In our pilot prospective fall study, 83% of the eligible pool of participants at the Knight ADRC agreed within 1 month of their clinical assessment to enroll in the study, and 98% agreed to be contacted for future fall studies.

DS supplement:

We will provide paper and electronic distribution of our flyer to patients within the clinical office of Dr. Ances at WUSM, along with presenting our study and distributing flyers to the following organizations: Down Syndrome Association of Greater St. Louis, United Cerebral Palsy- Heartland, St. Louis ARC, Albert Pujols Wellness Center for Adults with Down Syndrome, St. Louis Office for Developmental Disability Resources, Down Syndrome Innovations, and other community agencies throughout Missouri and the Metro East Region. We will also utilize Volunteer for Health and will post the recruitment flyer on social media. We will also utilize EPIC to identify potential candidates based on our eligibility criteria.

Consent

A study team member will approach all potential participants who meet the inclusion criteria and invite them to participate in the study. The informed consent form will be administered to all patients interested in participating in the study.

During the consent process, participants will: (1) have what the study is about and what is expected of them explained in detail either over the phone, over Zoom/HIPAA secure platform, or in person when remote is not possible; (2) discuss potential problems that could interfere with participation either over the phone, Zoom/HIPAA secure platform, or in person when remote is not possible; (3) have their questions answered either over the phone, Zoom/HIPAA secure platform, or in person when remote is not possible; and (4) receive a summary of the study and contact information for the PI and study coordinator. Informed consent to participate in the study will be obtained before any test or measurements are performed. The consent form will be signed by a witness and will be stored in REDCap, or paper copies will be stored in the office of the PI under double locks. Participants will be advised in the consent form that there is a possibility that their medical research record, including identifying information, may be inspected and photocopied by officials of federal or state government agencies and the Washington University Human Research Protection Office (HRPO).

For potential participants who have computer access and capability, the formal study consent process will be conducted using a REDCap-based electronic consent form. The consent form has been developed in REDCap, a secure, Web-based, HIPAA-compliant data collection platform with a user management system allowing the PI to grant and control varying levels of access to study staff. Potential participants would receive an email with a unique link to review the informed consent form online. After the research team explains the study and answers any questions, the potential participants can electronically fill in an “Agree” button, followed by their electronic signature. There is also a timestamp in REDCap that is utilized for the e-consent process. It is turned on and active, and each consent will have a date and timestamp for the signature. Upon completion of the consent, participants are presented with the option to download a copy of the executed form. The research team will also email a copy of the executed form to the participant. E-consent versioning will be managed using the e-consent framework in REDCap. Within the e-consent survey options, we have designated the e-consent version number in this application as e-consent version 1. The PDFs of completed responses will have the timestamp, participant name, and e-consent version number inserted in the footer. Future versions of the e-consent will be created by making a copy of the REDCap form and revising it. The old version would be deactivated upon receiving institutional review board (IRB) approval for the new version.

If a participant does not have access to a computer or smart device at the time of consent, potential participants will be provided with a copy of the informed consent in person or via mail. Once the potential participant has had time to look over the consent form, a study team member will talk with the participant by phone to review the study information and answer any questions. If they choose to continue their participation, they will be asked to sign the consent and return it to us either by email (scan or photo) or regular mail (postal service). The signed consent form will then be downloaded (if applicable) and retained by research staff. We will instruct participants to keep a copy for themselves.

We will continue to use the electronic consent process even after COVID restrictions have been lifted. All procedures will be the same as described above but with an option for signing in person using the electronic consent on a study-provided iPad. This will happen as follows: Once the potential participant has had time to look over the consent form provided to them either via email or paper copy, a study team member will talk with the participant in person to review the study information and answer any questions. If they decide to participate, they will be asked to sign the electronic version of the consent using a study-provided iPad. We will email a copy of the consent form if the participant has a valid email address; otherwise, we will download a copy and provide it to the participant. In the event that participants who consented using the REDCap e-consent would need to be re-consented, we will send the participant a link to the new version to discuss, sign electronically, download, and receive via email as described above.

The following privacy protections will be enacted for all patient communication via email: (1) a test email will be sent to the participant to verify their identity (confirm correct recipient) and that this email will be sent in a secure manner (i.e., [secure] in subject line); (2) the body of the email will instruct the participant to send all

information as a response to this thread and to not remove the "[secure]" from the subject line; (3) we will document in our research records that the participant agreed to provide information over email.

Starting in Year 2, participants who enrolled prior to the approval of the November 21, 2019 IRB modification will be re-consented to participate in all updated assessments and procedures; however, monthly phone calls will continue as part of the original consent.

DS supplement:

During the consent process we will use a stepwise approach (screening followed by consent) through the MacArthur Competency Assessment Tool for Clinical Research⁴¹ which is the gold standard of assessing capacity to participate in research within populations with potential decreased decision-making abilities. In this process we will include: Screening-1) Presentation of initial consent-simplification and 2) understanding and assent; Enrollment- 3) continual knowledge dissemination of the study process and procedures, 4) periodic discussions of decisional capacity throughout the study with the participant and/or caregiver to ascertain if there is continual understanding and assent (Appendix A).

Primary Intervention – N/A

Coronavirus (COVID-19)

An update about the study and study-related visits during the pandemic will be sent to all participants who have enrolled as of March 17, 2020. Participants will continue to receive automated calls or emails, but in-home visits are postponed. Participants who are due for their annual in-home visit may complete the questionnaires/assessments via phone, email, and/or Zoom/HIPAA secure telehealth platform during this time. They will also be re-consented to participate in updated procedures and assessment as needed. If they agree to participate in the stool portion, they will be mailed a stool kit.

Data Collection

Currently collected assessments at the Knight ADRC

Many measures of interest are captured as part of annual clinical examinations,⁴² neuropsychological evaluations,⁴³ and biomarker acquisition⁴⁴ conducted at the Knight ADRC.

Knight ADRC clinical assessments: Demographic variables (age, sex, race, APOE ε4, and CDR),³⁹ and depression will be collected and included as covariates.³⁷ The annual clinical assessment battery is conducted by experienced clinicians using a standardized protocol. The evaluation includes a health history, medication and depression inventories, and a general physical and neurologic examination of the participant.^{37,45,46} As we are enrolling participants at the time of their clinical assessment, we will use the assessment closest to enrollment as the baseline assessment. Demographics including age, sex, race, and educational attainment are collected or confirmed per a standardized protocol. *The Clinical Dementia Rating (CDR)* is obtained during the annual clinical examination. Clinicians interview the participant and a collateral source using a standardized protocol. The CDR is based on performance in six domains: memory, orientation, judgment and problem solving, community affairs, home and hobbies, and personal care. The instrument has established reliability.⁴⁷ CDR scores are assigned using a standardized algorithm and range from 0 (cognitive normality) to 3 (severe dementia). In individuals with CDR >0, the diagnosis of AD dementia and other dementing conditions is made in accordance with standard criteria based on the principle of intra-individual change.⁴⁸ The *Geriatric Depression Scale Short Form (GDS-SF)*⁴⁹ will be used to assess depression, a known fall risk factor.⁵⁰ Specifically designed and validated for older adults,⁵¹ the 15-item GDS-SF correlates highly with the original 30-item test ($r = .84$). Per GDS-SF scoring guidelines, we will classify a score of 5 or more as indicative of probable depression. *ε4 (APOEε4) genotype* is the strongest risk factor for AD but is also associated with poor mobility and physical function.^{37,52} We will include APOE ε4 genotype as a covariate. APOEε4 genotyping is performed using polymerase chain reaction on DNA isolated from whole-blood samples as described previously.³⁷ Participants who have one or more APOEε4 allele(s) will be identified as carriers.

Knight ADRC psychometric assessment: Participants will complete a 2-hour psychometric battery within ~2 weeks of their annual clinical assessment. Trained psychometrists blind to the participant's preclinical AD status will administer the tests via a standardized protocol.⁴³ The battery is composed of standard paper-and-pencil measures and novel computerized measures. Our battery has sound psychometric characteristics that have previously been shown to accurately differentiate healthy aging from preclinical AD.⁵³ We have identified patterns of cognitive performance that are highly predictive of the transition from healthy aging to AD, including autopsy-verified AD.⁵⁴ Our center was among the first to demonstrate that declines in cognitive functioning are associated with abnormal levels of *in vivo* AD biomarkers, including amyloid imaging⁵⁵ and CSF biomarkers including amyloid, tau, and p-tau.⁵⁶ Our novel computerized measures of attentional control have shown sensitivity to preclinical and early-stage symptomatic AD using reaction time distributional analyses,^{56,57} analyses of errors and switch costs,⁵⁸⁻⁶⁰ variability in controlled timing performance,⁶¹ and semantic retrieval.⁶² We have completed extensive factor and latent growth curve analyses of the existing Knight ADRC database and have developed cognitive composites. These cognitive composites include two memory composites (including episodic), two attentional control composites, a visuospatial composite, and a global cognitive composite.

Brain neuropathology assessments: To detect the *in vivo* presence of amyloid in the brain, PET using the radiotracer [18F] Florbetapir are obtained from participants within the Knight ADRC.⁶³ Soluble levels of amyloid ($A\beta_{42}$) are obtained from CSF studies. To detect the presence of neurofibrillary tangles, t-tau and p-tau are assayed from CSF.^{64,65} To detect neurodegeneration, brain volumetrics will be analyzed for cortical and subcortical signature areas affected by AD.⁶⁶ **PET imaging analysis:** Florbetapir images are obtained using a standard protocol on a 3T Siemens Biograph mMR hybrid scanner that allows for simultaneous MRI and PET acquisition. This reduces the burden on subjects and improves registration between imaging modalities. Quantitative image analysis is performed using a standard amyloid imaging analysis protocol⁴⁴ that uses FreeSurfer regions of interest (ROIs). Regional standard uptake value ratios (SUVRs) are obtained using the cerebellum as the reference region. **CSF biomarkers:** CSF analyses ($APOE_{42}$, t-tau, and p-tau₁₈₁; Innostest, Innogenetics: Ghent, Belgium) are collected according to a standard procedure⁶⁷ using sensitive and quantitative enzyme-linked immunosorbent assays (ELISAs). CSF is obtained using a 22-gauge Sprotte spinal needle to draw 20–30 mL of CSF at 8:00 AM following an overnight fast. CSF samples are gently inverted and centrifuged at a low speed to avoid possible gradient effects and are frozen at -84°C after being aliquoted into polypropylene tubes. All biomarker assays include a common reference standard, within-plate sample randomization, and strict standardized protocol adherence. Samples are re-analyzed if coefficients of variability (CVs) exceed 25% (per ADNI criteria), there are “edge artifacts” in the ELISA, there are suboptimal bead counts in xMAP, or reference standard yields widely discrepant values. **Brain volumetrics:** High-resolution structural MRI scans are acquired using a T1-weighted magnetization-prepared rapid gradient echo (MPRAGE) sequence. Images are subsequently analyzed using standard procedures developed at the Knight ADRC including FreeSurfer to delineate brain regions typically affected by AD.⁶⁶ **Preclinical AD staging**²⁰: The following cutoffs have been operationalized for biomarkers collected at the Knight ADRC²²: CSF <459 pg/mL for $A\beta_{1-42}$, >339 pg/mL for t-tau, and >67 pg/mL for p-tau₁₈₁; hippocampal and cortical signature volumes 1.5 standard deviations (SD) below the norm; and episodic memory composite score in the lowest 10th percentile. Participants will be classified as: **CN** if episodic memory and all CSF markers and brain MRI markers are normal; **Stage 1** if CSF $A\beta_{1-42}$ is abnormal or Florbetapir mean cortical binding potential is greater than 1.22; **Stage 2** if amyloid PET or CSF $A\beta_{1-42}$ and either CSF t-tau/p-tau₁₈₁, or brain volumetrics are abnormal; and **Stage 3** if amyloid PET or $A\beta_{1-42}$ and either CSF t-tau/p-tau₁₈₁, brain volumetrics, and episodic memory are abnormal. Participants will be classified as **SNAP** with no amyloid, elevated t-tau or p-tau, with or without subtle cognitive decline. SNAP individuals will be excluded.

Network dysfunction assessments: During the MRI scan, two rs-fc runs are obtained using a gradient spin-echo sequence. Participants are instructed to fixate on a visual cross-hair and not to fall asleep. Rs-fc pre- and post-processing is performed using standardized methods developed by our group.⁶⁸ In preparation for correlation analysis, data are spatially smoothed with a 6mm full-width at half maximum Gaussian blur. Then, temporal low-pass filtering ($f < 0.1$ Hz) is applied to the time series of each voxel. Finally, spurious variance is removed using linear regression for: (1) six parameters generated from head motion correction, (2) the whole

brain signal, and (3) signal from ventricular and cerebral white matter. An ROI-based analysis consisting of 298 seeds is performed with a Pearson's correlation coefficient computed between pairwise ROI time courses across all areas within RSNs. From these 298 seeds, correlation matrices are generated for each participant. For the 13 RSNs, correlation coefficients across ROI pairs within a network are averaged to form a composite score. Based on average matrices, both intra- (diagonal) and inter- (off diagonal) network composite scores are generated.

Central nervous system assessments (Table 2): We have identified patterns of cognitive performance in attention and executive control measures that are highly predictive of the transition from healthy aging to symptomatic AD,^{54,69} including autopsy-verified AD.⁵⁴ We will use sensitive composite scores of attentional and executive control tests (Table 2) to examine if it uniquely predicts functional mobility and falls.⁴³ In order to provide a strong test of this hypothesis, we will also generate a memory composite based on the Logical Memory Test⁷⁰, Selective Remind Tests,⁷¹ and the associate memory test, which has previously been shown to be sensitive to preclinical change.⁴³ In this way, we will be able to examine the degree to which the model fit is influenced by the attentional control composite above and beyond the memory deficits of AD.

Table 2. Central and peripheral nervous system measures by construct

	Construct	Measure	Description
Central nervous system	Attentional/executive control composite derived	Stroop color naming task ^{72*}	Color naming of congruent (e.g., red), neutral (e.g., deep), or incongruent word (e.g., blue).
		Simon task ^{73*}	Naming direction of an arrow with a keypress that is spatially consistent or inconsistent with the location of the arrow including congruent and incongruent positioning
		Attentional switching task ^{74*}	Switching every other trial between making odd-even decisions and consonant-vowel decisions on bivalent stimuli (e.g., B14)
Peripheral nervous system	Standing, balance, and vestibular function	Center of pressure path ^{75,76}	Center of pressure path will be measured using Balance Tracking System (BTrackS) in the following trials: single limb, eyes open; single limb, eyes closed; double limb, eyes open; double limb, eyes closed
	Lower extremity strength and function	30-Second chair stand test ⁷⁷⁻⁷⁹	A score below the norm will be considered indicative of decreased lower extremity strength and function
		Handheld dynamometer ⁸⁰	Minimal change in the peak torque value for lower extremity strength will be measured
	Vision	ETDRS ⁸¹	Visual acuity score; number of correct letters read
		Pelli-Robson ⁸²	Contrast sensitivity; letter-by-letter
	Sensation	Vibration ⁸³⁻⁸⁵ and Sharp	A tuning fork will be used test the big toes Sharp (arms and legs)

* Collected by the Knight ADRC at clinical exams

Note. Early Treatment Diabetic Retinopathy Study (ETDRS) test

New assessments

Peripheral nervous system abnormalities (Table 2): Peripheral abnormalities will be assessed by a board-certified neurologist during the annual physical and neurological exam at the Knight ADRC. The standard neurological examination evaluates cranial nerves (including vision), motor strength (including lower extremity strength), coordination, and balance.⁹¹ In addition, quantification of muscle strength will be obtained through handheld dynamometers.⁸⁰ This instrument records the force produced by loading through tension or compression. The minimal change in the peak torque value can be measured in units of weight such as pounds or kilograms.^{80,92} To assess postural instability, a balance board will be used to quantify the center of pressure (CoP).^{76,93} The CoP path and velocity will be measured in the following trials: single limb, eyes open; single limb, eyes open, dual task; single limb, eyes closed; double limb, eyes open; double limb, eyes open, dual task, double limb, eyes closed.

In-home functional mobility assessments

(Table 3): It is likely that motor and cognitive changes share a common underlying pathophysiology.^{17,94-96} In particular, gait variability, speed, and dual-task have demonstrated evidence as potential predictors of the development of MCI and AD^{94,95,97} and falls.⁹⁸⁻¹⁰⁰ Opal sensors will also be worn by participants during gait and balance testing (Performance Oriented Mobility Assessment [POMA] and Timed Up and Go (TUG) tests [TUG, TUG_{cog}, and TUG_{man}]). A certified occupational therapy (OT) rater blinded to the status of the CN participants will conduct all assessments.

Table 3. Functional mobility assessments

Construct	Measure	Description
Dynamic balance and mobility	POMA ⁸⁶	A task-oriented assessment of 9 balance tasks and 7 items to assess gait
Gait speed	TUG test ^{78,87}	Timed task of standing up, walking 3m, turning, walking back, and sitting down
Dual-task gait	TUG _{cog} ^{88,89}	TUG test while reciting serial 3s with subtractions from various points, 90/80/70
	TUG _{man} ⁹⁰	TUG test while carrying a glass of water

Note. Performance Oriented Mobility Assessment (POMA); The Timed Up and Go (TUG) test

Fall ascertainment: This study will provide unique information to this population by collecting prospective monthly fall reports. Retrospective fall reporting has demonstrated an underreporting of falls^{101,102}; prospective daily recordings with interview confirmation are recommended.¹⁰²⁻¹⁰⁴ All participants will use the gold-standard for fall reporting (daily calendar journal, monthly reporting, and fall interviews).¹⁰⁴ This technique was developed and tested in the Knight ADRC population.

Calendar journal: All participants will record daily falls using a tailored calendar journal.¹⁰⁵ The technique includes using daily habits to remember to record, monthly incentives, and landmark events on the calendar to improve recall. Monthly, participants will return journal pages via U.S. mail or report falls via automated call/email. Throughout the journal, spaces are available to record daily whether a fall or no fall occurred. A “fall hotline” is also available for participants who wish to report their fall immediately. Participants receive phone calls to prompt return of missing pages and/or to respond to the monthly call/email.¹⁰⁴ A **fall report form** is provided in the calendar so participants can note the time, location, nature of the fall environment, specific activity at the time of the fall, and any somatic complaints that proceeded the fall.¹⁰⁶

Fall interview: A fall report from the calendar will be followed up via telephone interview with a trained interviewer blinded to preclinical AD status.¹¹⁶ Falls, defined as unintentional movements to the floor, ground, or an object below knee level, will be verified.¹⁰⁴ Fall details will be ascertained, including injuries or medical treatment, using a standardized fall report form.^{27,105,117} Severity of injuries will be ascertained from medical records and participant report.¹¹⁸ We will determine the rate (number) and severity of falls (calculated with a standardized algorithm).¹⁰ **The fall severity score** will be quantified using a previously published algorithm: no falls (0); one fall without serious injury (1); any fall with minor injury or more than one fall (2); and major injury requiring hospitalization (3).⁹

Fall covariates assessments:

Recognized fall covariates^{119,120} will be assessed at the Knight ADRC and may be completed over the phone, via email, and/or in home at baseline and annual follow-up (Table 4). Fall risk assessments were chosen using the criteria for practical physiological assessments¹²⁰ (Table 3). Falls have multifactorial risks,¹²⁰⁻¹²² and the majority of falls in older adults occur at home.¹²³ In-home, real-world assessments contribute to the fall risk profile by adding factors that cannot be assessed in a clinical setting.

Factors such as dependence in ADLs¹²⁴ have been associated

with an increased fall risk. Intrinsic factors that can affect falls include incontinence,¹⁰⁹ pain, depression, vision,¹²⁵⁻¹²⁷ fear of falling,^{128,129} and alcohol abuse.¹³⁰ Extrinsic factors such as polypharmacy^{131,132} and home hazards¹³³ are also associated with falls. Opal sensors will also be worn by participants during the Westmead Home Safety Assessment. A fall risk composite score will be calculated using established cut-off values. Scores will be dichotomized as 0 for no fall risk and 1 for fall risk (see Table 3 for cut off scores) and summed to create a composite fall risk score.¹⁰⁷ The composite will be used to assess annual overall fall risk and as a covariate in analyses.

Table 4. Fall covariate composite score variables

Construct	Measure	Description	Fall risk cut-off ¹⁰⁷
Vision	ETDRS test ⁸¹	Visual acuity score; number of correct letters read	≤6/12
	Pelli-Robson test ⁸²	Contrast sensitivity; letter-by-letter	<36 letters
Alcohol abuse	SMAST-G ¹⁰⁸	10-item interview	≥2
Urinary incontinence	Frequency and type ¹⁰⁹	Short questionnaire of type (stress, urge, or other) and frequency	≥weekly urge incontinence
Depression*	GDS-SF ⁴⁹	15-item questionnaire; 0-15 points	>4
Pain	Self-report ¹¹⁰	Pain scale from SF-12	≥moderate
Medication*	Medication review	Medications and dosages	≥4 medications
Functional capacity	OARS ADL scale ¹¹¹⁻¹¹³	Ability to perform 14 activities; 0-2 scale, higher scores indicate greater independence	>4
Previous falls	Previous falls ¹⁰⁷	Total falls in past 12 months	>0
Home hazards	Westmead Home Safety Assessment ¹¹⁴	Rates 72 environmental home hazards as hazard/no hazard	≥4 hazards
Self-efficacy	FES-ISF ¹¹⁵	7 daily activities; rated from 1 (not at all) to 4 (very concerned) about falling during the activity	>10

*Collected by the Knight ADRC at clinical exams
 Note. Early Treatment Diabetic Retinopathy Study (ETDRS) test; Short Michigan Alcoholism Screening Test Geriatric Version (SMAST-G); Geriatric Depression Scale (GDS-SF); Older Adult Resources Services Activity of Daily Living (OARS ADL); Falls Efficacy Scale-International Short Form (FES-ISF)

Additional assessments of interest are listed in Table 5.

Table 5. Additional assessments

Construct	Measure	Description
Depression	PHQ 9 ¹³⁴	10-item questionnaire to assess frequency of symptoms; 0–27 points
Grip strength	Handheld dynamometer ¹³⁵	Pounds of force will be captured for grip strength
Functional performance	PASS ¹³⁶	Evaluates independence, safety, and adequacy with shopping, checkbook balancing, and medication management.
Falls behavior	FaB ¹³⁷	30-item questionnaire; rated from 1 (least protective) to 4 (most protective) behaviors to prevent falls
Immune dysfunction	Stool sample	Stool sample and associated dietary and gastrointestinal questionnaires
Olfaction	UPSIT ¹³⁸	40-item smell identification test; 0-40 points
Hearing	HHIE-S ¹³⁹	10-item questionnaire to screen for hearing impairment; 0-40 points.
	Brief Hearing Test	Screening tone test at varying frequencies

Note. Patient Health Questionnaire (PHQ 9); Performance Assessment of Self-Care Skills (PASS); Falls Behavioral Scale for Older People (FaB); University of Pennsylvania Smell Identification Test (UPSIT); Hearing Handicap Inventory for the Elderly Screening Version (HHIE-S)

Stool: Participants will be invited to provide optional stool samples around the time of the annual in-home assessment. If they choose to participate, participants will complete questionnaires regarding gastrointestinal function and dietary intake. They will also report any updates to their medications and health status (i.e., any recent illnesses, hospitalizations, etc.) each month (CDR ≤ 2 only in the stool collection portion of the study report prn). A subset of participants will be asked if they are willing to provide additional samples to examine the stability of the gut microbiome over time.

Coronavirus (COVID-19) assessments: In response to the coronavirus (COVID-19) pandemic, we will ask participants about their access to technology, willingness to participate in telehealth visits as needed, and COVID-19 diagnoses and symptoms. Participants will also have the opportunity to complete additional questions each month by phone or email. (Table 6).

Table 6. Coronavirus (COVID-19) assessments

Construct	Measure	Description
COVID-19 Experiences & symptoms	Have you been diagnosed with covid-19? y/n Have you been social distancing because of covid-19? y/n	
Alcohol/drug use	Have you increased your alcohol (including beer, wine, or liquor) or drug use since January 2020 (before the coronavirus pandemic began)? y/n	
Financial impact	HCP Financial Impact	11-items; measures financial stress related to the pandemic
Instrumental support	NIH Toolbox Instrumental Support Scale ^{140,141}	8-item questionnaire; rated from 1 (never) to 5 (always), higher scores indicate more support
Physical & mental health	PROMIS Global Health ¹⁴²	10-item questionnaire to assess general perceptions of health
Perceived stress	NIH TB Perceived Stress CAT 18+	10-items, rated on a 5-point Likert scale
Resilience	CD-RISC 10 ¹⁴³	10-items; 0–40 points, higher scores indicate greater resilience
Loneliness	UCLA Loneliness Scale ¹⁴⁴	20-item questionnaire; higher scores indicate more loneliness
Social isolation	PROMIS Social Isolation ¹⁴⁵	4-items; ranked from 1 (never) to 5 (always), lower scores indicate less social isolation
Depression	PHQ 9 ¹⁴⁶	10 items to assess frequency of symptoms; 0–27 points
Anxiety	PROMIS Anxiety (4 item) ¹⁴⁷ HADS ¹⁴⁸	4 items ranked on a 5-point Likert scale, higher scores indicate more anxiety
Sleep	PSQI ¹⁴⁹	9-item questionnaire; scale of 0-3, higher scores indicate “poor” sleeper

Note. Patient Reported Outcomes Measurement Information System (PROMIS) Global Health; 10-Item Connor-Davidson Resilience Scale (CD-RISC 10); Patient Reported Outcomes Measurement Information System (PROMIS) Social Isolation; Patient Reported Outcomes Measurement Information System (PROMIS) Anxiety Short Form; Hospital Anxiety and Depression Scale (HADS); Pittsburgh Sleep Quality Index (PSQI)

DS supplement:

Research Goal 1: To assess fall risk and functional mobility among adults aging with DS. Clinical assessments of fall risk covariants and functional mobility/performance: We will utilize existing participant data from the ABC-DS study at baseline or collected within the in-home assessment related to fall risk factors,¹⁵⁰ which include: hearing,^{7,151} orthostasis,¹⁵² vision,^{153,154} upper extremity (UE) strength,⁶ incontinence,¹⁵⁵ pain,¹⁵⁶ depression,¹⁵⁷ sensation,¹⁵⁸ motor coordination and balance,¹⁵⁹ and polypharmacy^{159,160} (see Table 7). We will also use the Dementia screening questionnaire for individuals with intellectual disabilities¹⁶¹ to qualitatively collect information regarding cognition and daily function of individuals with DS.

Table 7. DS supplement assessments			
Construct	Measure	Description	Risk cut-off
Hearing**/**	Hearing Handicap Inventory for the Elderly-screener (HHIE-S) ¹⁵¹	10-item questionnaire to screen for hearing impairment	≤8 no auditory concerns
	Caregiver general questions	2 questions assessing hearing difficulty	Profound hearing deficit
Orthostatic hypotension**/**	Blood pressure with change in position	1 and 3 min after standing	Systolic decline ≥20mmHg
Vision**/**	Visual Acuity ¹⁵³	Examination in ABC-DS clinic OR Documentation of last eye exam	ABC-DS study criteria or in home documentation
Sensation**/**	Peripheral Neuropathy Assessment ¹⁵⁸	Loss of Sensation questions only	Chronic loss of sensation
UE Strength**/**	ABC-DS clinical exam/ Dynamometer ¹⁶²	Muscle strength	ABC-DS study information/in home
Functional mobility**/**	Tinetti Performance Oriented Mobility Assessment ⁸⁶	A task-oriented assessment of 9 balance tasks and 7 items to assess gait	≤19 total score
Cognition**	Dementia screening questionnaire for individuals with intellectual disabilities ¹⁶¹	53 items and 3 parts; observer-rated questionnaire	≥20 total score, further testing advised
Medication**/**	Medication review	Medications and dosages	≥4 prescriptions
Depression**	Participant or caregiver report ¹⁵⁷	Depression & medication, 1 (yes), 0 (no)	depression 1, medication 0
Urinary incontinence**	Participant or caregiver report ^{21,109}	Short questionnaire of type, frequency, and incidents	≥weekly incontinence
Muscle Strength (LE)***	30 second Chair Stand Test ⁷⁷⁻⁷⁹	A score below the norm is indicative of decreased lower extremity strength and function	varies by age and gender
Gait Speed***	Timed up and Go ^{78,87}	Timed Task of standing up, walking 3m, turning, walking back and sitting down	TUG ≥ 12 sec
Pain**	Pain Likert Scale** PEG-3: Pain Screening Tool ¹⁶³	Visual Pain Likert Scale	≥7 pain
Functional capacity***	Waisman Activities of Daily Living Scale (W-ADL) ¹⁶⁴	Ability to perform 17 activities	>4 dependent ADL
Previous Falls /Fear of Falling (FOF)/Self Efficacy**	PROFANE Guidelines – Self/Caregiver Report ¹⁶⁵	5 questions about falling	varies by fall severity and history of falls
	Tinetti Fall Efficacy Scale-international ¹¹⁵	7 daily activities; rated from 1 (not at all) to 4 (very concerned) about falling during the activity	≥10
Home hazards***	Westmead Home Safety Assessment ¹¹⁴	Rater checklist of 72 fall home hazards	>4 hazards
* Note. *ABC-DS data, **Participant or Caregiver-report, ***Performance based assessment			

Research Goal 2: To investigate the prevalence of falls in aging adults with DS, and when falls occur within the amyloid, tau, and neurodegeneration (ATN) model of AD. Evaluation of fall ascertainment: participants with assistance from caregivers, or caregivers only, will fill out a monthly fall calendar and to report any falls monthly for 3-9 months.

Research Goal 3: To determine a power estimation for a future standardized multisite study. Data management, power calculation, and statistical analyses: The data obtained from participants for this study will include: (1) demographic information and medical history, (2) Functional performance data and the proposed study measures, and (3) fall calendar documentation. The data will be entered into Research Electronic Data Capture (REDCap)¹⁶⁶ and verified, and statistical programming will be completed in SAS.¹⁶⁷

Statistical Analyses and Sample Size Calculations

Data will be entered into REDCap¹⁶⁶ and verified. Programming will be done in SAS.¹⁶⁸ Differences in baseline characteristics across groups will be compared using appropriate statistics (chi-square test, Student's t-test, or

Mann–Whitney U test). The fall severity score will be quantified using a previously published algorithm: no falls (0); one fall without serious injury (1); any fall with minor injury or more than one fall (2); and major injury requiring hospitalization (3).⁹ Preclinical AD staging²⁰: The following cutoffs have been operationalized for biomarkers collected at the Knight ADRC²²: CSF <459 pg/mL for A β _{1–42}, >339 pg/mL for t-tau, and >67 pg/mL for p-tau₁₈₁, and episodic memory composite score in the lowest 10th percentile. Participants will be classified as CN if both episodic memory and all CSF markers are normal, Stage 1 if only A β _{1–42} is abnormal, Stage 2 if A β _{1–42} and either t-tau or p-tau₁₈₁ are abnormal, and Stage 3 if A β _{1–42} and either t-tau or p-tau₁₈₁ and episodic memory are abnormal. Participants will be classified as SNAP with no amyloid or elevated t-tau or p-tau, with or without subtle cognitive decline. Individuals with SNAP will not be included for further analysis.

Aim 1a correlates falls and functional mobility with AD biomarkers to determine whether falls and functional mobility are associated with preclinical stages of AD.

Statistical Analysis Plan for Aim 1a. Analyses to address Aim 1a will be cross sectional, using falls over 1 year and baseline functional mobility (POMA, Timed up and Go [TUG]) across the preclinical stages of AD.²⁰ We will first examine the distributions of falls, POMA, and TUG dual-task across four preclinical stages: 0, 1, 2, and 3. If the distributions are not normal, we will seek appropriate transformations (e.g., Box-Cox transformations¹⁶⁹). Subsequent analyses will be conducted on transformed values. Falls (number and severity) and functional mobility across preclinical stages will be compared through group means or adjusted means from the original or the transformed scale using analysis of covariance (ANOCOVA) models.¹⁷⁰ In addition to preclinical stages of AD, ANOCOVA models for falls will adjust for other covariates including age, depression, sex, fall risk composite score, and APOE ϵ 4 genotype, as well as possible interactions between these factors and preclinical stages. These analyses will be implemented by PROC GLM/SAS.¹⁶⁸ Similar analyses will be conducted to determine whether functional mobility measures are also associated with preclinical stages of AD. A major analytic concern in these tests is the control of overall type I error rate due to multiple outcome variables (number and severity of falls, functional mobility) tested in this aim. We will address this concern via two different approaches. First, we will implement a Monte Carlo algorithm and partition testing procedure, such as Hochberg's method (1988)¹⁷¹ and Holm's method.¹⁷² Second, we will seek composite markers (i.e., those based on factor analysis and principal component analyses) from all of these functional variables and will conduct primary analyses using these composite markers.

Power Analysis for Aim 1a. We will enroll 350 older (≥ 65 years old) participants from the Knight ADRC. Based on existing data concerning the Knight ADRC cohort (see preliminary data),²² we estimate that 41.5% will be at preclinical Stage 0, 15% at Stage 1, 12% at Stage 2, and 4% at Stage 3. We powered Aim 1a by testing the difference of fall (number and severity) and functional mobility variables across the four preclinical AD stages. Assuming the effect size (i.e., mean difference) for a functional composite outcome between any two adjacent participant groups (from Stage 0 to 3) is the same, the proposed sample sizes will provide at least 80% statistical power to detect an effect size as small as 0.225 SD (same SD assumed across the four preclinical stages) on the composite outcome between two adjacent participant groups. Our power calculation is based on an F-test (with three degrees of freedom for the numerator) on the main effect for each of the preclinical stages with a one-way ANOVA model at a significance level of 5%.

Aim 1b determines whether change in preclinical disability falls (number and severity) and functional mobility (POMA, TUG) over time predicts conversion to AD (operationalized by clinical and cognitive measures). We hypothesize that an increase in falls and functional mobility impairments will be associated with a change in global cognitive function and will predict conversion to symptomatic AD.

Statistical Analysis Plan for Aim 1b. The primary analysis for Aim 1b is to associate longitudinally measured falls with the risk of subsequently developing symptomatic AD. To achieve this, we will jointly model the longitudinal falls (number and severity) and the time to the symptom onset of AD (defined as the first time a participant receives a CDR >0 due to AD). For modeling the longitudinal rates of changes in number and severity of falls, we will use general linear mixed effects models.¹⁷³ If a linear trend fits the data over time, we will use standard random intercept and slope models.¹⁷⁴ Otherwise we will seek nonlinear models such as the piecewise linear model.¹⁷⁵ For modeling the risk of developing AD, we will use the semi-parametric Cox proportional hazards model. To address the association between change in falls and the risk of developing symptomatic AD, we will incorporate the latent random intercept and slope from the general linear mixed model

into the Cox proportional hazards model in the joint analyses, which will allow us to test the hypothesis that the rate of change in falls predicts the development of AD. More specifically, we will implement similar joint models as in Jacqmin-Gadda et al.¹⁷⁶ and De Gruttola & Tu.¹⁷⁷ Because death is a potential competing risk in the progression of dementia, two “survival times” will be observed for each participant: the time from baseline to death, and the time from baseline to the clinical assessment when the participant has a CDR >0 and with an etiologic diagnosis of AD. We will use the marginal models approach, which includes three common variants¹⁷⁸ to model the risk of disease progression as a function of baseline biomarker values. These marginal models amount to fitting individual ordinary semi-parametric Cox’s proportional hazards models¹⁷⁹ for each “survival time” and then using the robust sandwich variance/covariance estimator to account for the correlational structure between two different “survival times” from the same participants. These approaches have been used in similar biomedical applications.¹⁷⁹ We will also examine the effect of potential confounding covariates such as baseline age, depression, sex, fall covariate composite, and APOE ε4 status and conduct further analyses after adjusting for the effects of these covariates. We will compute and report the asymptotic 95% CIs for the HR associated with the disability scores. Model goodness-of-fit will be assessed through various model diagnostic tools such as the residual plots and formal statistical tests by introducing time-dependent covariates.¹⁸⁰ Similar longitudinal analyses will also be conducted to determine whether longitudinal changes in functional mobility (POMA, TUG) measures are also predictive of the conversion to symptomatic AD.

Power Analysis for Aim 1b. We calculated the power to test the primary scientific hypothesis that faster increases in falls over time are associated with an increased risk of progression from CN to symptomatic AD. We used the current Knight ADRC database to fit a survival curve from baseline to the time that a CDR >0 was first rendered and found that the estimated rate of CDR progression is ~7.2% per year for those with a mean age of ~75 years old at baseline, similar to the cohort in this proposal. Based on these estimates and assuming similar rates of attrition we have experienced in the past, we estimate that out of 350 CN older adults who will be assessed at baseline, at least 300 participants will continue to be assessed annually for CDR and cognition during the 5 years of the grant with an average follow-up time of ~4 years. We predicted that 75 individuals will progress to CDR >0 by the end of the funding cycle. This estimated number of individuals who will progress to CDR >0 after baseline within the 5-year funding period will provide at least 80% statistical power to detect a one-fold increase in the risk (e.g., HR of 2) of developing dementia for individuals with a fast increase in falls over time (defined by the worst 50% on the rate of change) in comparison to those with slow or no change in falls over time (defined by the best 50% on the rate of change). These power computations were based on a log rank test at a 5% significance level and assumed an annual rate of 4.7% of CDR progression for individuals with slow changes in disability over time.

Aim 2 examines the potential mechanisms for falls in preclinical AD using rs-fc.

Statistical Analysis Plan for Aim 2. We will conduct exploratory analyses to compare rs-fc in the CON between the CN older adults with and without falls using a Student’s t-test or a Wilcoxon rank-sum test, depending on the distribution of rs-fc. In addition, we will control for the age effect, preclinical stages of AD, and other covariates (e.g., depression) using analysis of covariance (ANCOVA). We will also compare domain-specific and global cognitive measures, amyloid, and neurodegenerative measures between CN older adults with and without falls using the Wilcoxon rank-sum test.

Power Analysis for Aim 2: Our other rs-fc studies suggest that at least 80% of subjects will participate in rs-fc studies. We estimate that a total of 280 subjects will have at least baseline rs-fc assessments. Based on a Student’s t-test, this sample size provides at least 80% power to detect a difference of 0.35 SD in rs-fc composite scores between individuals with and without falls. This power analysis assumed 60% of out participants will have falls, again based on the estimate from our own database.

DS supplement:

Statistical Analysis of Goal 1: To assess fall risk and functional mobility among adults aging with DS. The differences in baseline characteristics will be explored using appropriate statistics (i.e., chi-square test, Student’s t test, or Mann-Whitney U test).¹⁸¹ Differences in baseline characteristics across groups based on demographic variables (age, gender, and race if applicable) will be compared using Fisher’s exact test for categorical data.

Statistical Analysis of Goal 2: To investigate the prevalence of falls in aging adults with DS. We will

explore the distributions of fall severity scores among all participants. If the distribution is not normal, we will seek appropriate transformations (e.g., Box-Cox).⁴⁹ Subsequent analyses will be conducted on transformed values. Falls severity will be explored in DS participants through group means (adjusted or transformed) using analysis of covariance (ANOCOVA) models. ANOCOVA models for falls will be adjusted for other covariates including the biological variable of sex, a fall risk composite score, and amyloid precursor protein (APP) on chromosome 21 (obtained by the ABC-DS study), as well as possible interactions among these factors and any identified preclinical stages of AD (from the ABC-DS neurological data and parent study data if applicable).

Statistical Analysis of Goal 3: To determine a power estimation for a future study. We will utilize the variance of functional mobility and fall risk measures within our sample to estimate the magnitude of a clinically significant difference between the measures, with an avoidance of a type 1 error. Our plan is also to determine the necessary number of subjects needed to detect an effect between the fall and functional performance outcomes and their relationship to preclinical AD.

Data Sharing

DS supplement:

De-identified data will be shared on an as needed basis according to the NIH data sharing policies. Data will be shared using a controlled access database. De-identified data will be shared with, and received from, Dr. Amy Bodde and Dr. Brian Helsel at The University of Kansas Medical Center. The information that will be shared include the following: demographic information (age, gender, race/ethnicity, mild/moderate level of disability, health history [comorbid conditions], orthostatic hypotension/blood pressure, living status), and the Dementia Screening Questionnaire For Individuals With Intellectual Disabilities, Timed Up and Go, Waisman Activities of Daily Living Scale, Grip strength via handheld dynamometer, and Chair Stand Test. This information will be used to prepare manuscripts and conference presentations.

Strengths and Limitations

We have designed the study to have high levels of rigor including: (1) the scientific premise of falls during preclinical AD is explicitly justified using high-quality studies and our own preliminary data; (2) the study is designed to produce robust and unbiased results (e.g., blinded raters re-certified semi-annually, standard operating procedures, conservative sample size estimates, plan to collect data from dropouts); (3) relevant biological variables (e.g., sex) and important fall covariates are justified and included; and (4) all procedures for clinical and biomarker data collection are consistent with National Alzheimer's Collaborating Center (NACC)¹⁸² and Alzheimer's Disease Neuroimaging Initiative-2 (ADNI-2)¹⁸³ to ensure authentication. The study is designed to generate reproducible data (standardized protocols, assessments, and commercially available monitoring) and provide a foundation for future studies.

This study benefits from the existing infrastructure already in place at the Knight ADRC. The Knight ADRC provides a unique opportunity to monitor falls and functional mobility in a large, well-characterized cohort of CN older adults. Potential pitfalls. In anticipation of problems, we have designed several safeguards. For example: (1) Although we have planned for attrition, dropout from the Knight ADRC longitudinal studies could impact the findings of this study. To prevent missing data, we will train and certify all in-home OT raters to conduct the CDR in the home. If participants refuse clinical studies but agree to home assessment, they will be retained. (2) If older adults are not compliant with fall monitoring over time, we have piloted an automated phone response system module in REDCap to ascertain falls weekly or daily (depending on participant preference). The system is in use in our current trial and is immediately deployable. A response of fall immediately connects participants to the research coordinator. (3) It may be possible that there is not an increased risk of falling during the preclinical period of AD. It may be that falls are only a sensitive (not specific) predictor of preclinical AD. Understanding the pattern of behavioral change may not only be useful to understand preclinical AD, but can also be used as a comparator to the preclinical stages of other neurological disorders such as Parkinson's disease. Information on fall rates and mechanisms could be instructive for future treatment targets. (4) Falling is a syndrome highly associated with general aging and may be difficult to differentiate from aging-related phenotypes such as frailty. We will collect important covariates to test these relationships in a cohort of older adults without preclinical AD. It is possible that comorbid medical conditions, substance abuse, and other fall risk factors found in the general population could introduce bias. Therefore, we will control for these covariates in statistical analyses.

DS supplement:

This supplement also has high levels of rigor, including: (1) the scientific premise of falls in adults aging with DS is explicitly justified using high-quality studies generated by the parent study and B. Ances affiliation with the ABC-DS study; (2) the study is designed to produce robust and unbiased results (e.g., standard operating procedures, conservative sample size estimates, plan to collect data from dropouts); (3) relevant biological variables (e.g., sex) and important fall covariates are justified and included.

This supplement benefits from the existing infrastructure already in place at the Ances clinic and Stark Participation, Environment and Performance Laboratory. The Participation, Environment and Performance Laboratory provides a unique opportunity to monitor falls and functional mobility in a population of adults with Down syndrome. Potential pitfalls. In anticipation of problems, we have designed several safeguards. For example: (1) Although we have planned for attrition, dropout from the ABC-DS study or individuals enrolled within the community could impact the findings of this study. To prevent missing data, if aging adults with DS are not compliant with fall monitoring over time, we have piloted an automated phone response system module in REDCap to ascertain falls weekly or daily (depending on participant preference). The system is in use in within the parent study and is immediately deployable. (3) It may be possible that there is not an increased risk of falling during the 6-month observation period. Understanding the pattern of behavioral functional change may not only be useful to understand the pathway to preclinical AD, and the information on fall rates and mechanisms could be instructive for future study design. (4) It is possible that comorbid medical conditions, substance abuse, and other fall risk factors found in the general population could introduce bias. Therefore, we will control for these covariates in statistical analyses.

Potential Benefits, Risks, and Alternatives

Benefits

Participants will be provided with the results of the assessments conducted during their annual in-home visit. This study will provide researchers with a better understanding of the longitudinal changes that older adults experience during the pre-clinical course of AD. Advances in this area have the potential to improve quality of life for older adults and allow physicians and researchers to better identify older adults in the preclinical phases of AD. A better understanding of falls and functional mobility in individuals at risk for AD is needed for informing interventional studies.

DS supplement:

Advances in this area have the potential to provide physicians and researchers with a better understanding of falls and functional mobility in individuals aging with DS who are at risk for AD. This information may inform interventional studies and improve quality of life for aging adults with DS.

Risks

Potential risks of research participation—physical, psychological, financial, and legal risks among others—are considered minimal (45CFR46.404). In-home evaluations and assessments of barriers may result in fatigue or aggravation. In addition, some questions may touch on emotionally sensitive issues that could cause anxiety or other forms of emotional stress. The performance-based testing involves observation of everyday activities, which may result in fatigue or embarrassment. In rare instances, the evaluation protocol could result in a fall. Participants will be told that their involvement in this research study is voluntary and that they may choose not to participate or withdraw their consent at any time. Withdrawal from the study will not at any time affect the commitment of the clinician to administer care, and there will be no penalty or loss of benefits to which participants are otherwise entitled. Participants who undergo the study visits will be given the option to reschedule the visit or take a break at any time during the study if necessary. There is little legal risk to participating in this research. All research-related information will be kept confidential and accessible only to authorized members of the research team.

Minimization of Risks and Confidentiality

We are a HIPAA-covered entity and comply with all HIPAA regulations. To protect against and minimize potential risks, participants will be carefully screened and evaluated for eligibility by the research coordinator.

To avoid or minimize symptoms of fatigue, agitation, or emotional distress due to testing, participants will be instructed to notify the rater or interventionist if they experience any discomfort. They will also be periodically questioned about their tolerance for the tests/intervention. Testing and interviews will be terminated if participants develop fatigue, agitation, or emotional distress. Risks during the home visit will be minimized by having trained interventionists available to monitor safety during the home assessment. When photographing the home, only environmental barriers and changes will be collected. Images of participants will not be collected.

An ID number will be assigned to each participant. All data collected from a participant will be labeled with the ID number. All participant electronic and hard-copy data will be kept under double-lock protection. All hard-copy forms that contain personal identifiers (e.g., name, address, phone numbers) will be stored in a separate locked file drawer under double-lock protection. No publication or presentation of the study data will uniquely identify or provide sufficient information to uniquely identify participants.

To guard against unauthorized data access, all shared-use computer systems at Washington University School of Medicine are protected with passwords, which are changed at 4-month intervals. Only individuals with a particular "need to know" status are given access, and system privileges are carefully restricted. All personal computers to be used in the Administrative Unit are located within a secure area, and the system is locked when not in use. SAS and SPSS software packages will be used for data management and analysis. Datasets generated from these programs will be password protected, which will make accessing study data difficult even in the event of unauthorized computer access. Systems connected to the Ethernet are carefully controlled, and all systems without Ethernet access control are insulated from the backbone by bridges or routers. The Ethernet cable itself is routed only through secure passageways.

Data are directly entered into a REDCap database.¹⁶⁶ REDCap servers are securely housed in an onsite, limited-access data center managed by the Division of Biostatistics at Washington University. All Web-based information transmission is encrypted. All data are stored on a private, firewall-protected network. All users are given individual user IDs and passwords, and their access is restricted on a role-specific basis. REDCap was developed specifically around HIPAA security guidelines and is implemented and maintained according to Washington University guidelines. Study data will be collected via tablet in the field and managed using REDCap electronic data-capture tools hosted at Washington University. REDCap is a secure, Web-based application designed to support data capture for research studies.

Adverse Event Reporting and Safety Monitoring

All serious adverse events (SAEs) will be reported to the HRPO in the following time frames: (a) death immediately; (b) life-threatening, within 7 calendar days; (c) all other SAEs, within 15 calendar days using the electronic serious adverse event reporting system. Should there be an SAE that occurs that increases the risks to the participants, the study will be stopped, an investigation will be conducted, and a findings report will be generated before the study is resumed.

Dr. Stark will be responsible for reviewing study progress and outcomes including recruitment, data quality, and safety. Quarterly reports will be reviewed by the study investigators. Because risk in the proposed study is considered minimal, the data monitoring plan will include continuous, close monitoring by the study investigator with prompt reporting of any adverse events. In this study, Dr. Stark will monitor the study for adverse events and adherence to the protocol and safety.

Indemnity

Washington University School of Medicine is responsible for any non-negligent damage incurred as a result of participating in the trial. The indemnity is renewed on an annual basis. Washington University School of Medicine assures that it will continue renewal of the indemnity for the duration of the trial.

Ethics and Dissemination

This protocol and the template informed consent forms will be reviewed and approved by the Washington University IRB with respect to scientific content and compliance with applicable research and human subjects regulations. All study personnel involved in the conduct of this research will receive the required education on the protection of human participant rights.

Authorship Eligibility and Contributorship

Authorship for this study will be given to key personnel involved in study design, recruitment, data collection, and data analysis. There are no publication restrictions, and no professional writers will be involved in the generation of the manuscript. S. Stark and B. Ances were responsible for conceptualizing study design. S. Stark managed patient safety protocol and IRB compliance. S. Stark and B. Ances were responsible for recruitment, enrollment, data collection, and editing the protocol.

DS supplement:

S. Washington, R. Bollinger, S. Stark and B. Ances will be responsible for conceptualizing study design. S. Washington and R. Bollinger will manage patient safety protocol and IRB compliance. S. Washington, R. Bollinger, S. Stark, and B. Ances will be responsible for recruitment, enrollment, data collection, and editing the protocol.

All authors agree to be accountable for the accuracy and integrity of all aspects of the trial.

Funding**Competing Interests**

None of the authors have conflicts of interest to disclose.

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Appendix A.

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Capacity assessment outcomes and plans

A ABLE

- Researcher may obtain consent
- Recommend the researcher remind the subject that if he later changes his mind about participating that decision will be respected

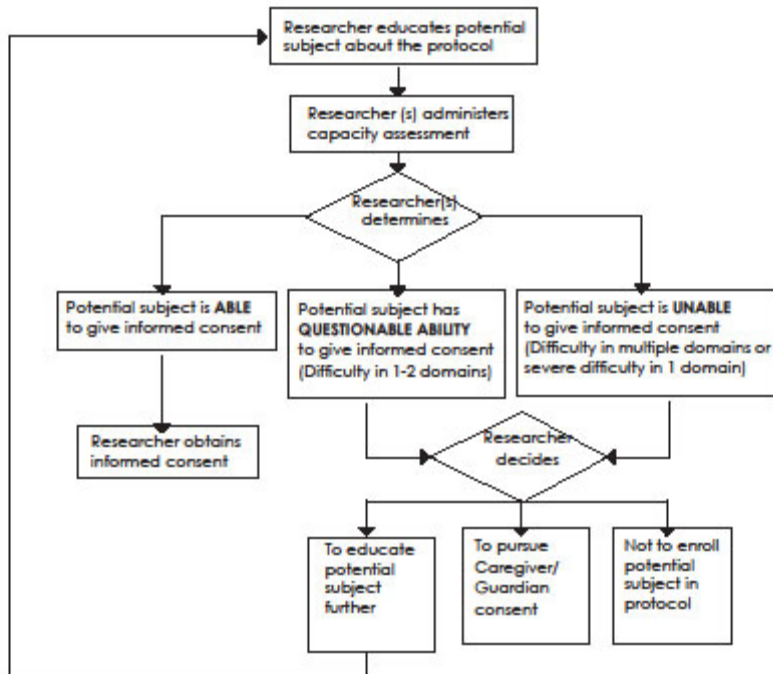
B QUESTIONABLE ABILITY

- Recommend further education in domains where difficulty was noted
- Re-assess

C UNABLE

- Researcher may pursue surrogate consent as allowed in this protocol
- Assess ability to assign a surrogate decision-maker
- Assess appropriateness of the surrogate

Capacity assessment algorithm



Note: MacArthur Competency Assessment Tool for Clinical Research⁴¹ ; National Institute of Mental Health Human Subjects Protection Unit (HSPU)¹⁸⁴